Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		A black body [is a body or surface which] absorbs all the [electromagnetic] radiation that falls upon it / no body is a better emitter of radiation [at any wavelength] than a black body at the same temperature	1			1		
	(b)	(i)	Valid strategy e.g. IR^2 = constant (1) Data from graph used to validate relationship e.g. $(2 \times 10^{11})^2 \times 0.8 = 3.2 \times 10^{22}$ $(4 \times 10^{11})^2 \times 0.2 = 3.2 \times 10^{22}$ (1) Award both marks for correct use of data only			2	2	2	
		(ii)	Correct substitution of corresponding pairs of values into $I = \dfrac{P}{4\pi R^2}$ regardless of units used (1) Correct re-arrangement and correct unit conversions to show clearly that $P \approx 4 \times 10^{26}$ W e.g. $P = 1.4 \times 10^3 \times 4\pi \times (1.5 \times 10^{11})^2$ (1)	1	1		2	2	
	(c)		λ_{peak} found from graph (= $500 \pm 10 \times 10^{-9}$) [m] (1) Wien's law to find T_{sun} i.e. $\dfrac{2.9 \times 10^{-3}}{500 \times 10^{-9}}$ [= 5 800 K] (1) (**ecf** on λ_{peak}) Substitution into $P = 4\pi R^2_{sun}\sigma T^4$ (**ecf** on T) e.g. $4 \times 10^{26} = 4 \times \pi \times R^2_{sun} \times 5.67 \times 10^{-8} \times (5\,800)^4$ (1) $R_{sun} = 7.0 \times 10^8$ m (1) **unit mark**	1	1 1 1		4	4	
			Question 7 total	**3**	**4**	**2**	**9**	**8**	**0**